Chapter 16

Optimal Taxation with Commitment

Ramsey Model — Primal Approach

1. Environment

The government goal is to maximize households welfare subject to raising prescribed revenues through **distortionary taxation** (taxes that affect economic decisions). In the simplest version, the production factors are raw labor and physical capital on which the government levies distortionary *flat-rate* taxes.

Key features of the model:

- The government takes into account the response of the private sector to taxation and debt.
- The government can **credibly commit** to future taxes.
- If an equilibrium has an asymptotic steady state, then the optimal policy is eventually to set the tax rate on capital to **zero** (Chamley–Judd result).

2. Household

A continuum of measure one of infinitely-lived consumers with lifetime utility:

$$\sum_{t=0}^{\infty} \beta^t \big[u(c_t) + v(\ell_t)\big], \qquad \beta \in (0,1) \tag{1}$$

where $u(\cdot)$ is increasing, strictly concave and three-times continuously differentiable; $v(\cdot)$ is decreasing. The household is endowed with one unit of time:

$$\ell_t + n_t = 1 \tag{2}$$

Intuition: *The family wants to be as happy as possible along its life (infinite), the happiness depends on the consumption c_t and the leisure ℓ_t. Moreover $u(c_t)$ is the utility of consumption and $v(\ell_t)$ the utility for labor. $v(\cdot)$ is decreasing so more labor implies less happiness. β is the discount factor, the patience, how you value today. It is decreasing over t. If β is low, you care more about today. If β is high (close to 1) you don't care too much between the future and today.*

Budget constraint

$$c_t + k_t \;=\; w_t\,\phi\,\ell_t\,(1-\tau_t^{\ell}) \;+\; k_{t-1}\big[1 + (r_t-\delta)(1-\tau_t^{k})\big] \tag{3}$$

Symbol	Definition
w_t	Wage
ϕ	Labor productivity $(\phi = 1)$
τ_t^{ℓ}	Proportional labor income tax
τ_t^{k}	Proportional capital income tax
r_t	Interest rate
δ	Capital depreciation rate
g_t	Exogenous government expenditure

3. Firm

The representative firm produces using capital and effective labor:

$$F(K_{t-1},\, e_t) \tag{4}$$

The firm maximizes profits:

$$\max_{K_{t-1},\, e_t} \; F(K_{t-1}, e_t) - r_t K_{t-1} - w_t e_t$$

Intuition: *The firm takes two principal ingredients, the capital from 1 period in the past, and the effective labor, and wants to maximize its profit function.*

Firm optimality conditions (FOCs)

$$F_1(K_{t-1}, e_t) = r_t \tag{5}$$

$$F_2(K_{t-1}, e_t) = w_t \tag{6}$$

Each factor is paid its marginal product (perfect competition).

Intuition: *The marginal productivity of capital is equal to the rent of capital r. If you produce x extra, you pay x in rent. The marginal productivity of labor: the extra production from a worker is equal to the wage. Perfect competition: each factor gets exactly what it contributes, there are no extra benefits.*

4. Market Clearing Conditions

$$\phi\,\ell_t = e_t \qquad \text{(Labour market)}$$

$$K_t = k_t^g + k_t \qquad \text{(Capital market)}$$

$$c_t + g + K_t - (1-\delta)K_{t-1} = F(K_{t-1}, e_t) \qquad \text{(Goods market)}$$

Initial conditions: K_{-1} and k_{-1}^g given.

Intuition: *The production must be enough to cover the consumption, the amount that the government needs, and the replenishment of depreciated capital.*

5. Characterising Competitive Equilibria

Euler equation

$$u'(c_t) \;=\; \beta\, u'(c_{t+1})\big[1 + (r_{t+1} - \delta)(1 - \tau^k_{t+1})\big]\,, \qquad \forall\, t \tag{7}$$

The marginal utility of consuming today must equal the marginal utility of consuming tomorrow, adjusted by the discount factor β and the net return on capital after taxes.

Intuition: *The marginal utility of consuming today must equal the marginal utility of consuming tomorrow times the saving plus the interest return after taxes, times the discount rate.*

Consumption–labor optimality condition

$$-\frac{v'(\ell_t)}{u'(c_t)} \;=\; w_t\, \phi\, (1 - \tau^\ell_t) \tag{8}$$

Intuition: *Each extra unit of labor makes you "unhappy", $v(\cdot)$ is decreasing so $v'(\cdot) < 0$. The marginal utility is concave. Marginal labor / marginal consumption is like happiness/labor (hour) / happiness/consumption (euro) so it's analogous to euros/hour.*

6. Primal Approach: Deriving the Implementability Constraint

Goal: rewrite the policy problem in terms of *allocations only*, eliminating prices (r_t, w_t) and taxes (τ^ℓ_t, τ^k_t).

Intuition: *Now working with the primal approach, the logic to build the implementability condition and eliminate prices and taxes is that the government prefers to find the allocations of capital, consumption and labor that generate the needed taxes, not set the taxes and then look for the correct allocations. The implementability condition is built from the budget constraint, the labor-consumption optimality condition, and the Euler equation. Once the government has the consumption and the labor, it can recover the needed taxes.*

Step 1 — Substitute the consumption–labor FOC into the budget constraint

From the consumption–labor condition:

$$w_t\, \phi\, (1 - \tau^\ell_t) \;=\; -\frac{v'(\ell_t)}{u'(c_t)}$$

Substituting into the household budget constraint:

$$c_t + k_t \;=\; -\frac{v'(\ell_t)}{u'(c_t)}\, \ell_t \;+\; k_{t-1}\big[1 + (r_t - \delta)(1 - \tau^k_t)\big]$$

Multiplying both sides by $u'(c_t)$:

$$u'(c_t)\,c_t \;+\; v'(\ell_t)\,\ell_t \;+\; u'(c_t)\,k_t \;=\; u'(c_t)\,k_{t-1}\big[1+(r_t-\delta)(1-\tau_t^k)\big] \tag{9}$$

Step 2 — Use the Euler equation to eliminate the tax term

Writing the Euler equation shifted one period back (from $t-1$ to t):

$$u'(c_{t-1}) \;=\; \beta\,u'(c_t)\big[1+(r_t-\delta)(1-\tau_t^k)\big]$$

Solving for the term that appears on the right-hand side of Step 1:

$$u'(c_t)\big[1+(r_t-\delta)(1-\tau_t^k)\big] \;=\; \frac{u'(c_{t-1})}{\beta}$$

Substituting back:

$$u'(c_t)\,c_t \;+\; v'(\ell_t)\,\ell_t \;+\; u'(c_t)\,k_t \;=\; k_{t-1}\,\frac{u'(c_{t-1})}{\beta} \tag{10}$$

Step 3 — Multiply by β^t and sum over all periods

Multiplying both sides by β^t:

$$\beta^t u'(c_t)\,c_t \;+\; \beta^t v'(\ell_t)\,\ell_t \;+\; \beta^t u'(c_t)\,k_t \;=\; \beta^{t-1}u'(c_{t-1})\,k_{t-1}$$

Summing from $t=0$ to ∞:

$$\sum_{t=0}^{\infty}\big[\beta^t u'(c_t)\,c_t+\beta^t v'(\ell_t)\,\ell_t+\beta^t u'(c_t)\,k_t\big] \;=\; \sum_{t=0}^{\infty}\beta^{t-1}u'(c_{t-1})\,k_{t-1} \tag{11}$$

Step 4 — Telescoping: only endpoints remain

Separating the k_t term on the left and expanding the right-hand side:

$$\sum_{t=0}^{\infty}\beta^t[u'(c_t)c_t+v'(\ell_t)\ell_t] \;+\; \underbrace{\sum_{t=0}^{\infty}\beta^t u'(c_t)\,k_t}_{A} \;=\; \frac{u'(c_{-1})}{\beta}\,k_{-1} \;+\; \underbrace{\sum_{t=1}^{\infty}\beta^{t-1}u'(c_{t-1})\,k_{t-1}}_{B}$$

Writing out the terms of A and B explicitly confirms they are identical (both equal $\sum_{t=0}^{\infty}\beta^t u'(c_t)k_t$), so they cancel from both sides:

$$\sum_{t=0}^{\infty}\beta^t[u'(c_t)\,c_t+v'(\ell_t)\,\ell_t] \;=\; \frac{u'(c_{-1})}{\beta}\,k_{-1} \tag{12}$$

Note: the telescoping cancellation is exact term by term. Formally, the convergence of the infinite sum requires the ***transversality condition*** $\lim_{T\to\infty}\beta^T u'(c_T)k_T = 0$, *which is a necessary optimality condition for the household's problem and ensures the household neither accumulates unbounded assets nor dies in debt.*

Step 5 — Apply the Euler equation at $t = 0$

Evaluating the Euler equation at $t = 0$:

$$u'(c_{-1}) \;=\; \beta\, u'(c_0)\big[1 + (r_0 - \delta)(1 - \tau_0^k)\big]$$

Substituting into the right-hand side:

$$\frac{u'(c_{-1})}{\beta}\, k_{-1} \;=\; \frac{\beta\, u'(c_0)\,\big[1 + (r_0 - \delta)(1 - \tau_0^k)\big]}{\beta}\, k_{-1} \;=\; u'(c_0)\, k_{-1}\big[1 + (r_0 - \delta)(1 - \tau_0^k)\big]$$

Result: Implementability Constraint

$$\boxed{\sum_{t=0}^{\infty} \beta^t \big[u'(c_t)\, c_t + v'(\ell_t)\, \ell_t\big] \;=\; u'(c_0)\, k_{-1}\big[1 + (r_0 - \delta)(1 - \tau_0^k)\big]} \tag{13}$$

This single equation, expressed entirely in terms of **allocations**, summarises all the optimising behaviour of the private sector. Prices and taxes have been eliminated. Once the government finds the optimal allocations (c_t, ℓ_t, K_t), the corresponding taxes can be recovered from the household optimality conditions.

Note: the term τ_0^k on the right-hand side is ***not*** *a choice variable for the government — it is an* ***initial condition****, given at $t = 0$ alongside k_{-1}. The government only optimises over taxes from $t \geq 1$ onward.*

7. The Ramsey Problem

Tax limit every period (in primal approach)

From the household Euler equation:

$$u'(c_t) = \beta u'(c_{t+1})\left(1 + (r_{t+1} - \delta)(1 - \tau_{t+1}^k)\right)$$

Rearranging:

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1 = (r_{t+1} - \delta)(1 - \tau_{t+1}^k)$$

Solving for τ_{t+1}^k:

$$\frac{\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1}{r_{t+1} - \delta} = 1 - \tau_{t+1}^k$$

$$\tau_{t+1}^k = 1 - \frac{\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1}{r_{t+1} - \delta} \leq \tilde{\tau}$$

Now solve for $u'(c_t)$:

$$1 - \frac{\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1}{r_{t+1} - \delta} \leq \tilde{\tau}$$

$$-\frac{\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1}{r_{t+1} - \delta} \leq \tilde{\tau} - 1$$

Assuming $r_{t+1} - \delta > 0$, multiply by $-(r_{t+1} - \delta)$, so the inequality reverses:

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} - 1 \geq (1 - \tilde{\tau})(r_{t+1} - \delta)$$

$$\frac{u'(c_t)}{\beta u'(c_{t+1})} \geq 1 + (1 - \tilde{\tau})(r_{t+1} - \delta)$$

Therefore,

$$u'(c_t) \geq \beta u'(c_{t+1}) \left[1 + (r_{t+1} - \delta)(1 - \tilde{\tau})\right]$$

This is the tax-limit constraint written in primal form.

The Lagrangian

The Ramsey problem is subject to:

- the feasibility constraint,
- the implementability constraint,
- the tax-limit constraint.

The Lagrangian can be written as:

$$\begin{aligned}
\mathcal{L} = \sum_{t=0}^{\infty} \beta^t \Big\{ & u(c_t) + v(l_t) \\
& + \mu_t \Big[F(k_{t-1}, e_t) + (1-\delta)k_{t-1} - k_t - c_t - g_t \Big] \\
& + \Lambda \Big[u'(c_t)c_t + v'(l_t)l_t \Big] \\
& + \gamma_t \Big[u'(c_t) - \beta u'(c_{t+1})\big(1 + (r_{t+1} - \delta)(1 - \tilde{\tau})\big) \Big] \Big\} \\
& - \Lambda u'(c_0)k_{-1}\big(1 + (r_0 - \delta)(1 - \tau_0^k)\big)
\end{aligned}$$

Notice carefully that the last term

$$-\Lambda u'(c_0)k_{-1}\big(1 + (r_0 - \delta)(1 - \tau_0^k)\big)$$

is **outside** the summation braces.

Intuition: *Lagrangian multipliers:*

- μ_t: *how much happiness improves with an extra unit of product.*
- Λ: *how much happiness improves if the distortion from taxes reduces.*
- γ_t: *how much happiness improves if the government has more flexibility with the capital tax.*

Why impose a restriction on τ_0^k?

To make the Ramsey problem economically meaningful, we usually impose a restriction on the initial capital tax τ_0^k, for example by taking it as exogenously given and small. This rules out confiscatory taxation of the initial capital stock, that is, a capital levy that would act like a lump-sum tax.

FOC with respect to k_t

We now differentiate the Lagrangian with respect to k_t.

Step 1: Where does k_t appear?

k_t appears in three relevant places:

1. In the feasibility constraint at time t, through the term $-k_t$.
2. In the feasibility constraint at time $t+1$, through

$$F(k_t, e_{t+1}) + (1-\delta)k_t.$$

3. In the tax-limit term at time t, because

$$r_{t+1} = F_k(k_t, e_{t+1}).$$

Step 2: Derivative of each part

From the period-t feasibility term:

$$\frac{\partial}{\partial k_t}\left[\beta^t \mu_t \big(F(k_{t-1}, e_t) + (1-\delta)k_{t-1} - k_t - c_t - g_t\big)\right] = -\beta^t \mu_t$$

From the period-$(t+1)$ feasibility term:

$$\frac{\partial}{\partial k_t}\left[\beta^{t+1} \mu_{t+1} \big(F(k_t, e_{t+1}) + (1-\delta)k_t - k_{t+1} - c_{t+1} - g_{t+1}\big)\right] = \beta^{t+1} \mu_{t+1} \big(F_k(k_t, e_{t+1}) + 1 - \delta\big)$$

From the tax-limit term, since $r_{t+1} = F_k(k_t, e_{t+1})$ implies $\frac{\partial r_{t+1}}{\partial k_t} = F_{kk}(k_t, e_{t+1})$:

$$\frac{\partial}{\partial k_t}\left[-\beta^{t+1}\gamma_t u'(c_{t+1})\big(1 + (r_{t+1} - \delta)(1-\tilde{\tau})\big)\right] = -\beta^{t+1}\gamma_t u'(c_{t+1})(1-\tilde{\tau})F_{kk}(k_t, e_{t+1})$$

Step 3: Combine all terms

Adding all contributions and dividing by β^t:

$$\boxed{-\mu_t + \beta\mu_{t+1}\big(F_k(k_t, e_{t+1}) + 1 - \delta\big) - \beta\gamma_t u'(c_{t+1})(1-\tilde{\tau})F_{kk}(k_t, e_{t+1}) = 0}$$

8. Steady-State Analysis

Setting up the steady state

In steady state:

$$\mu_t = \mu_{t+1} = \mu_{ss}, \qquad \gamma_t = \gamma_{ss}, \qquad c_t = c_{t+1} = c_{ss}, \qquad F_k^{ss} \equiv F_k(k_{ss}, e_{ss}), \qquad F_{kk}^{ss} \equiv F_{kk}(k_{ss}, e_{ss})$$

The FOC with respect to k_t becomes:

$$\mu_{ss}\Big[-1 + \beta(F_k^{ss} + 1 - \delta)\Big] - \beta\gamma_{ss}u'(c_{ss})(1-\tilde{\tau})F_{kk}^{ss} = 0$$

Using the Euler equation in steady state

From household maximization:

$$u'(c_{ss}) = \beta u'(c_{ss})\Big(1 + (r_{ss} - \delta)(1 - \tau_{ss}^k)\Big)$$

By the Chamley–Judd result, in the long run $\tau_{ss}^k = 0$, so the Euler equation reduces to:

$$1 = \beta\big(1 + (r_{ss} - \delta)\big) \qquad \Longleftrightarrow \qquad 0 = -1 + \beta(F_k^{ss} + 1 - \delta)$$

using $r_{ss} = F_k^{ss}$ from perfect competition.

Intuition: *When $\tau^k = 0$, the Euler equation reduces to $1 = \beta(1 + r_{ss} - \delta)$. This means the net return on capital exactly compensates the household's impatience. What you have today equals what you will have tomorrow (which is more money), but given the impatience, you are indifferent.*

Final implication for γ_{ss}

Substituting into the steady-state FOC, the first bracket vanishes and we obtain:

$$-\beta\gamma_{ss}u'(c_{ss})(1-\tilde{\tau})F_{kk}^{ss} = 0$$

Assuming $\beta > 0$, $u'(c_{ss}) > 0$, $(1-\tilde{\tau}) > 0$, and $F_{kk}^{ss} \neq 0$:

$$\boxed{\gamma_{ss} = 0}$$

Thus, in the long run, the tax-limit constraint is **no longer binding**, because the optimal capital tax converges to zero.

Intuition: *When γ is non-active ($= 0$), the capital tax in the long run is strictly smaller than the limit set by the government for the capital tax. In this case, the tax is zero, which is the farthest possible distance from any limit.*